farm animals possess great economic value. Since prehistoric time human beings taken maximum advantage from animals by keeping them under their control through domestication. The economic success of the industries, based on animals and their products, depends on the proper production, management and development of the next generation of farm animals.

13.1 Scope of Zoology

Studying Zoology can provide self employment opportunities and you can become an entrepreneur. Economic Zoology is a branch of science that deals with economically useful animals. It involves the study of application of animals for human welfare. The need of Zoology is not just to improve our economic condition but also to provide food security and provide employment opportunities. Based on the economic importance, animals can be categorized as:

1. Animals for food and food products
2. Economically beneficial animals
3. Animals of aesthetic importance
4. Animals for scientific research

13.2 Vermiculture

Vermiculture is the process of using earthworms to decompose organic food waste, into a nutrient-rich material capable of supplying necessary nutrients which helps to sustain plant growth. The aim is to continually increase the number of worms to have a sustainable harvest. The excess worms can either be used to expand a vermicomposting operation or sold to customers. Vermicompost is the primary goal of vermiculture. Technically, the worm castings are pure worm waste and are fine and nutrient rich organic soil amendment. Vermicompost on the other hand, is comprised of the castings, bits of bedding and other organic matter. Essentially, though the terms are used interchangeably, they are both worm manure and are valuable for improving soil health. Applications of earthworm in technology of composting and bioremediation of soils and other activities is called Vermitech (Sultan Ismail, 1992).

The disposal of solid wastes (bio-degradable and non- biodegradable) remains a serious challenge in most of the countries. Earthworms play a vital role in maintaining soil fertility; hence these worms are called as **"farmer's friends"**. These are also called as **"biological indicators of soil fertility"**. The reason is that they support bacteria, fungi, protozoans and a host of other organisms which are essential for sustaining a healthy soil. The breakdown of organic matter by the activity of the earthworms and its elimination from its body is called vermicast. It is a finely divided granular material and is noted for its porosity, aeration, drainage and moisture holding capacity and serves as rich organic manure.

Earthworms are divided into two major groups. The first group, the **humus formers,** dwell on the surface and feed on organic matter. They are generally darker in colour. These worms are used for vermicomposting. The second group, the **humus feeders,** are burrowing worms that are useful in making the soil porous, and mixing and distributing humus through out the soil. There are different **endemic** (native) species of earthworms cultured

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